Objective of the Dashboard

The primary objective of the **Power BI E-Commerce Return and Refund Analysis Dashboard** is to help businesses track, understand, and control product returns and refunds to protect overall profitability. Specifically, its goals include:

- **Controlling Revenue Leakage**: Helping the business identify hidden financial leakages where money is being lost through customer refunds.
- **Explaining Return Patterns Quickly**: Resolving the client's problem where returns are visible in inventory, but the business cannot explain or diagnose them fast enough.
- **Isolating Root Causes**: Enabling decision-makers to pinpoint the exact product categories, subcategories, sales channels, or specific reasons driving the returns.
- **Evaluating Discount Risks**: Analyzing whether heavy discounts inadvertently drive low-intent purchases that result in a higher rate of product returns.

Key Performance Indicators (KPIs) Needed

The dashboard is split into two logical areas—the **Executive Summary** (Page 1) and **Diagnostic Design** (Page 2):

Page 1: Executive Summary KPIs

- **Total Orders**: A distinct count of unique Order IDs to understand overall transaction volume.
- **Return Orders**: The count of unique orders where the return flag is marked as "yes".
- **Return Rate (%)**: The proportion of total sales that were returned (Return Orders / Total Orders), representing key operational risk.
- **Refund Amount**: The total financial loss calculated dynamically by multiplying returned quantities by their unit price, while subtracting active customer discounts and refund deductions.
- **Return Units**: The sum of all returned product quantities.

- **Average Return Days**: Calculated via a date difference formula (DATEDIFF between delivery date and return date) to measure how long it takes for customers to return products.
- **Top Return Category**: Dynamically displays the product category experiencing the highest return volume (e.g., "Fashion").
- **Top Return Reason**: Dynamically displays the primary reason customers return items (e.g., "Product not as expected").

Graphs and Visualizations Needed

To satisfy the user's need for diagnostic depth and actionable insights, the dashboard incorporates several visual elements:

- **Monthly Trend Chart (Line and Stacked Column Chart)**: Combares the **Monthly Refund Amount** (primary Y-axis) against **Return Orders** (secondary Y-axis) over time. This helps identify seasonal spikes or trends in returns.
- **Return Units vs. Total Orders by Category (Clustered Bar Chart)**: Compares return units (highlighted in red) directly against total orders (highlighted in green) for each product category to evaluate relative return rates visually.
- **Sales Channel Distribution (Donut Chart)**: Slices return units by the channel of purchase (Marketplace, Mobile App, Website, or Offline Stores) to isolate high-risk avenues.
- **Category vs. Return Reason Heat Map (Matrix Table)**: Formatted with conditional background color gradients, this matrix lists subcategories in rows and return reasons in columns. It immediately spotlights major problem spots, such as *Men's Clothing* returns due to *"Wrong size"*.
- **Detailed Product Performance Table**: A clean table containing Product Name, Total Orders, Return Units, Return Rate %, Refund Amount, Average Return Days, and Average Discount for micro-level business analysis.

- **Slicers**:
    - **Date Slicer ("Between" Slider)**: To filter all visual metrics across any specific date range.
    - **Sales Channel Slicer**: To filter the summary dynamically by purchase touchpoints.
    - **Month Name Slicer (Dropdown)**: To inspect monthly performances.

Implementation steps create the dax :

Measure Name	DAX Syntax	Operational Description
Total Orders	Total Orders = DISTINCTCOUNT('packed orders'[Order ID])	Calculates the total unique order count across the synthetic database.
Return Orders	Return Orders = CALCULATE([Total Orders], 'packed orders'[Return Flag] = "Yes")	Isolates and counts only those orders that were physically returned.
Return Rate %	Return Rate = DIVIDE([Return Orders], [Total Orders], 0)	Computes the return frequency. After creating, select the measure and format it as **Percentage** under **Measure Tools**.
Refund Amount	Refund Amount = CALCULATE(SUMX('packed orders', 'packed orders'[Return Quantity] * 'packed orders'[Unit Price] * (1 - 'packed orders'[Discount Percentage]) * (1 - 'packed orders'[Refund Deduction Percentage])), 'packed orders'[Return Flag] = "Yes")	Calculates extremely precise financial leakage. It dynamically iterates row-by-row to subtract active discounts and refund deductions directly from raw transactional prices.